Does underlying structure predict quantum trainability and dequantization?

A dataset of 1 368 learning problems, and a first test of the structure we extract from them

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Can structure predict how a quantum model will behave?

1 Generic circuits do not train

McClean et al. 2018: in random, structure-free circuits, gradients shrink exponentially with the number of qubits (barren plateaus).

2 Structure restores gradients

Building the problem's symmetry in keeps the dynamical Lie algebra (DLA) small, and a small DLA gives large gradients (Larocca et al. 2022; Ragone et al. 2024).

3 ...and may remove the advantage

The same small DLA can let a classical computer imitate the model, i.e. dequantize it (Cerezo et al. 2025; Goh et al., g-sim).

Question. Can the underlying structure of a learning problem (symmetry, locality, spectrum, data complexity) predict, before training, whether structure-preserving quantum models of it will be trainable and resist dequantization? Only symmetry has a theorem linking it to the trainability formula; for the other three the link is our working hypothesis.

The project runs in two phases; this study checks the first links

Phase A

1

Collect problems

1 368 real and generated learning problems, four types

2

Extract structure

factor graph and 12 features per input; validated here

3

Build circuits

natural (twirled), generic control, path model with dial λ

4

Fingerprint & predict

DLA size and blocks; Ragone and g-sim predictions committed first

5

Measure

output variance, training outcome, classical simulation cost

Phase B: the predictor

A statistical model from a circuit's fingerprint to (trainable?, hard to simulate?). It is always tested on problem families it has never seen, and counts as a success only if it beats the theoretical formulas where their conditions fail: shallow circuits, approximate symmetry.

A problem is a learning task whose inputs define an operator

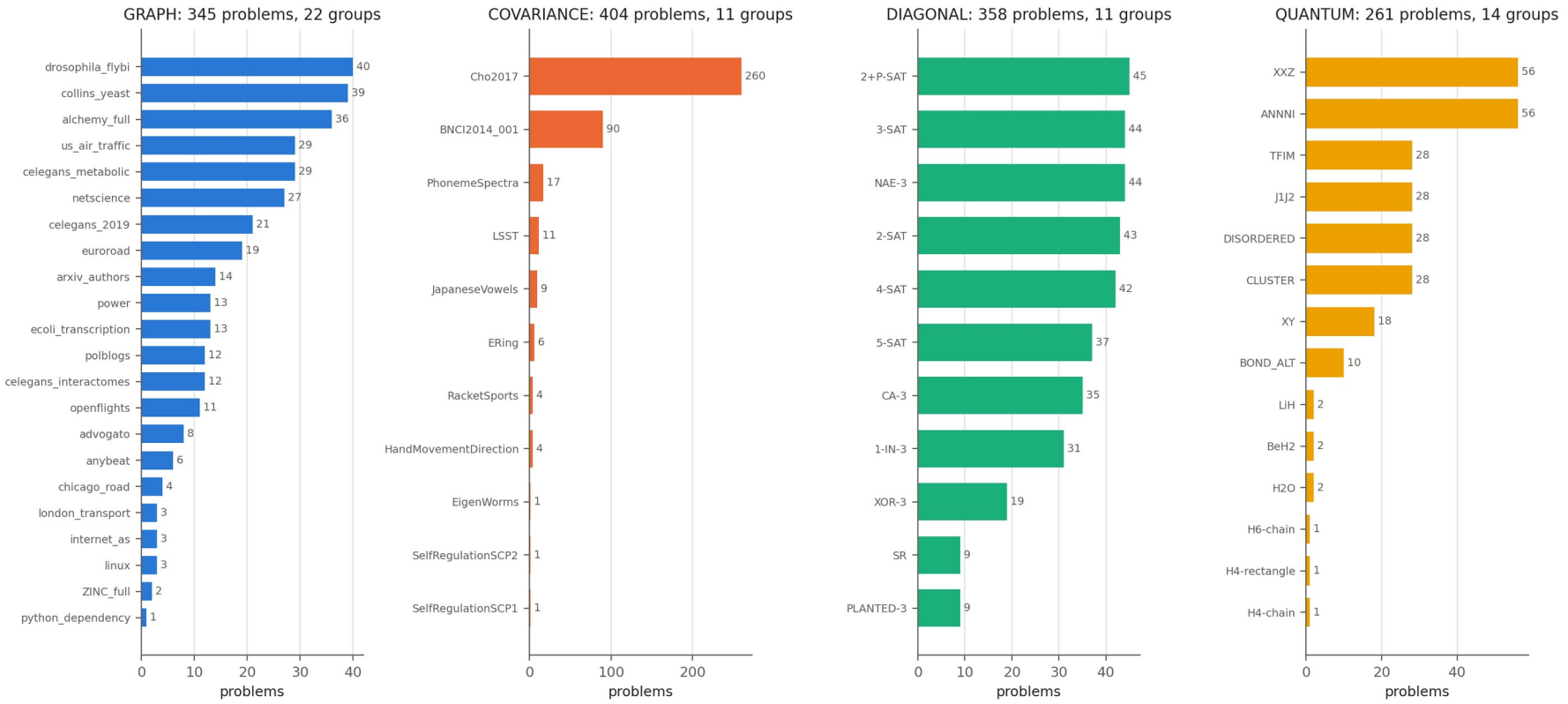
$$\mathbf{P_n} = (X_n, Y_n, \mu_n, E_n, f^*)$$

inputs on n = 4…12 sites, labels ±1, samples with a fixed 70/30 split, one differential equation per input through an operator M(x), and the label rule. The task operator $\bar{M}$ is built from TRAIN only.

Type	Operator M(x)	Equation	Sources
GRAPH	Laplacian $L = B^T W B$	heat	20 real networks (Netzschleuder), Alchemy & ZINC molecules
COVARIANCE	precision matrix $Q = R^{-1}$	Ornstein–Uhlenbeck	EEG (MOABB), 9 UEA sensor sets
DIAGONAL	clause-penalty Hamiltonian	Schrödinger	11 SAT generators
QUANTUM	non-commuting Hamiltonian	Schrödinger	8 spin chains, 6 molecules

The type is fixed by the operator, not by the domain: MaxCut encodes like a graph, 3-SAT has three-qubit terms a graph cannot have. Every input passes an equation check (0 failures).

The harvest: 1 368 problems from 58 independent sources



1 368

problems (binary tasks)

58

source groups: the true unit of independence

55%

of problems have real inputs (tiers 1–2)

Weak point: 350 of COVARIANCE's 404 problems come from 2 EEG datasets; GRAPH's 345 problems rest on 93 distinct input sets.

Cleaning: six stated rules, raw data untouched

Found in the raw harvest (1 379 problems)	Rule applied
TRAIN/TEST leak: identical inputs on both sides, 22 problems	R2 group-aware re-split; R3 weights
Class balance below 0.2 floor, 1 problem	R1 excluded
Degenerate ground states, 24% of spin samples	re-harvest in a fixed symmetry sector (642 → 10 on a dry run)
Hartree–Fock not converged, 2 H_2O geometries	R5 samples removed
Uninformative operator (identity), 475 UEA samples	R6 flagged, sensitivity check

22 → 0

problems with a TRAIN/TEST leak

0 FAIL

in the audit of the final set (2 WARN: TEST balance 0.17)

337 930

samples in 1 368 problems

The raw files' sha256 is identical before and after the build. XY-periodic (10 problems) excluded: rejected by the pipeline after a solver failure.

One common object for four types: the factor graph

Factor graph of M(x)

One node per site, one per term of the operator, each term joined to the sites it acts on and weighted by its normalised coefficient (Kschischang, Frey & Loeliger 2001). It keeps many-body terms, which a coupling matrix would lose.

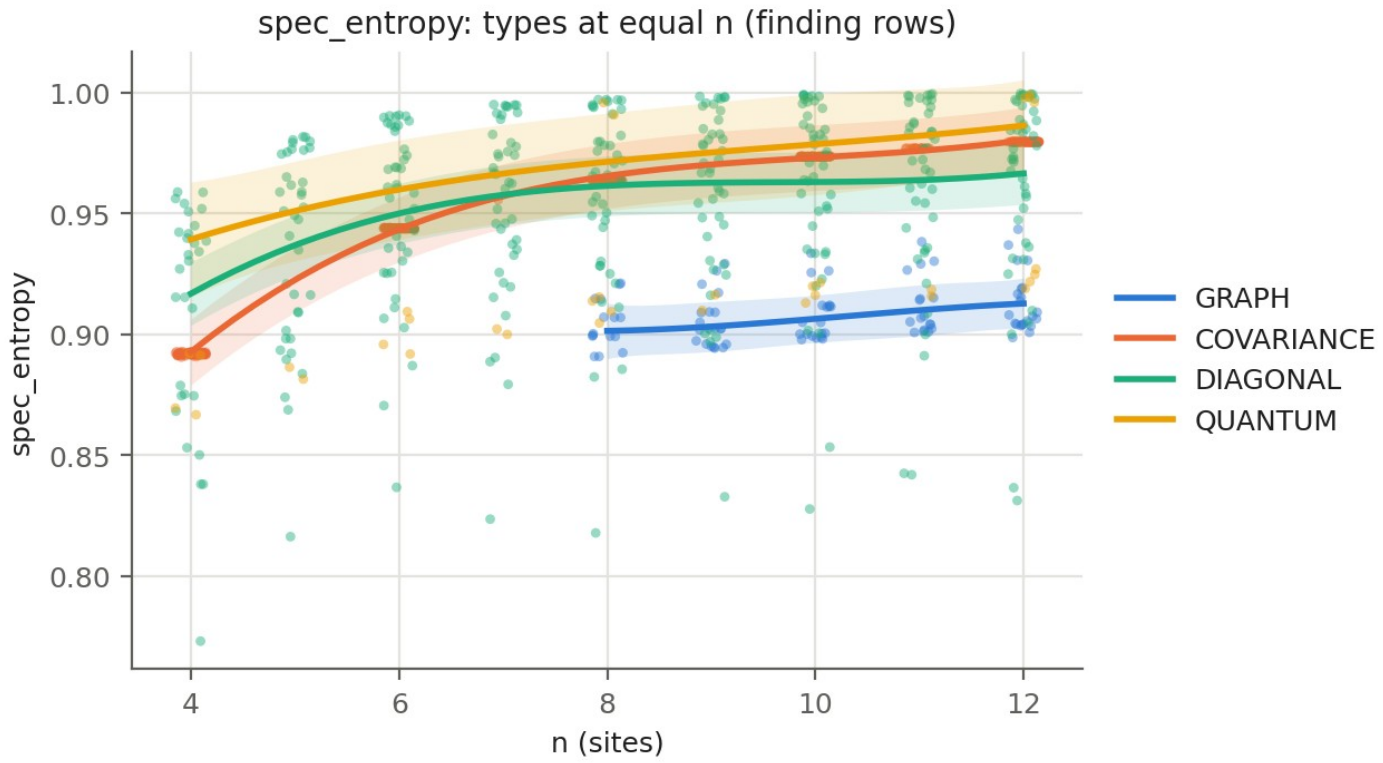
12 scale-free features

Locality: terms per site, term sizes, degree spread, support density. Symmetry: colour-refinement classes. Spectrum: entropy, participation ratio and connectivity of the factor graph's Laplacian. Median over each problem's inputs (to be revisited).

Status	Meaning	Use
forced	fixed exactly by the type's definition	pipeline check: must match exactly
predictable	follows from the generator's parameters	finding only if the data deviate
found	depends on real data	finding

Every (feature, type) cell was classified before any result was seen.

Q2: types differ in spectrum, but only GRAPH stands apart



Method. A mixed model per feature: type baseline + smooth curve in n per type + random offset per source (Wood 2017). Level and shape tests, Holm-corrected over 12 features.

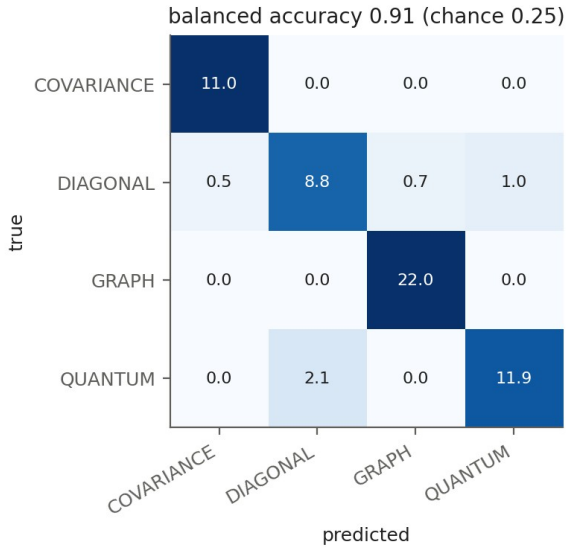
Result. Spectral features differ by type (p 10^{-12} to $5{\cdot}10^{-7}$) and scale differently with n; marginal R^2 0.30–0.55. GRAPH differs from every type ($p < 10^{-4}$); the other three are close (entropy at n = 10: 0.97, 0.96, 0.98).

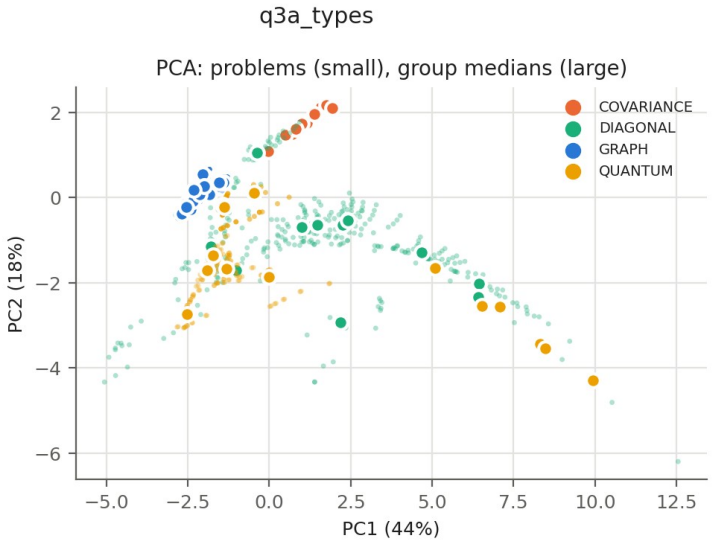
All forced values match exactly; results unchanged without flagged samples.

18–132

effective observations, not 1 107 (ICC 0.5–0.9): sources dominate

Q3: the type is recovered, the domain within a type is not

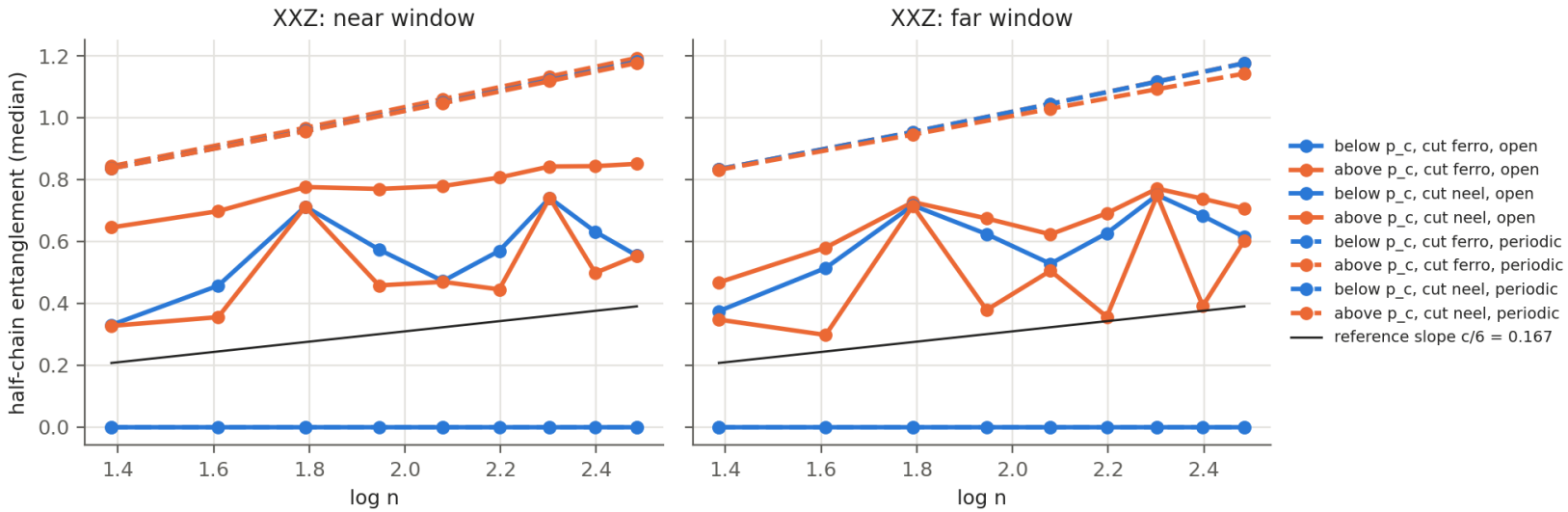




Task	Balanced acc.	Chance
Q3a: 4 types	0.91	0.25
Q3b: GRAPH domains	0.08	0.20
Q3b: COVARIANCE	0.49	0.50
Q3b: DIAGONAL	0.20	0.33
Q3b: QUANTUM*	1.00	0.50

Logistic regression, one source held out at a time, three size controls (n-only baseline, Dinga et al. 2020, size-balanced subset). Type errors fall where the operator definitions predict (NAE-3, 2-SAT are pairwise, read as GRAPH). Below-chance scores are a known effect of leave-one-group-out with few sources (Jamalabadi et al. 2016). *Trivial: forced spin-chain features.

Q4: structure changes lawfully with n



28 / 28

forced laws match exactly

379 / 428

trends monotone in n

19 / 177

laws nameable at n ≤ 12 (ΔBIC > 10)

Each sub-family followed across n = 4…12. On open XXZ chains in the critical phase, the entanglement slope is 0.18–0.21 against the Calabrese–Cardy prediction c/6 = 0.167, inside the interval in all four cases. Four readings depend on the flagged degenerate states.

What the study supports

1. **Correct.** every forced value and law matches; the one bug found was fixed before results were read
2. **Varied, but coarse.** spectral features separate GRAPH from the rest, not the other three types from each other
3. **Lawful in n.** monotone trends, Calabrese–Cardy on critical XXZ; most laws unnameable at $n \leq 12$
4. **Operator, not domain.** the kind of operator is recovered; the domain within a type is not
5. **Few independent sources.** 18–132 effective observations: favour source diversity, hold out whole sources

This validates the extracted structure; trainability and dequantization need simulated circuits.

Next steps

1 Finish the analysis

Run the permutation tests (Q2: 5 000, Q3: 1 000 group shuffles) and confirm the open analysis choices. The robustness study (Q6) was not done.

2 Summarise the inputs

Plot each problem's per-input distributions, then decide how to summarise them; align the molecules' origin tier across files.

3 Logical State Space

A first instance L = (C; F; Φ), Hodge Laplacians on each factor graph's complex: one medium to explain structure ⇒ function.

4 Then the rest of Phase A

Build natural, generic and path circuits from each problem, fingerprint them, commit the Ragone and g-sim predictions, and measure trainability and classical simulation cost; validate the predictor by holding out whole sources.

References

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Larocca et al., Group-invariant quantum machine learning, PRX Quantum 3, 030341 (2022)

Ragone et al., A Lie algebraic theory of barren plateaus, Nat. Commun. 15, 7172 (2024); Fontana et al., Nat. Commun. (2024)

Cerezo et al., Cost function dependent barren plateaus, Nat. Commun. 12, 1791 (2021)

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Goh et al., Lie-algebraic classical simulations for quantum computing, arXiv:2308.01432

Kschischang, Frey & Loeliger, Factor graphs and the sum-product algorithm (2001)

Wood (2017); Pedersen et al. (2019); Aarts et al. (2014); Nakagawa & Schielzeth (2013): mixed models

Roberts et al. (2017); Dinga et al. (2020); Jamalabadi et al. (2016): grouped validation

Calabrese & Cardy (2004); Hastings (2007); Kass & Raftery (1995): laws in n

Data: Netzschleuder; TUDataset (Morris et al. 2020), Alchemy (Chen et al. 2019); MOABB; UEA archive (Bagnall et al. 2018)